

OBERVELLACH, Austria

WMO: 112610

Lat: 46.929N

Lon: 13.223E

Elev: 2654

StdP: 13.34

Time Zone: 1.00 (EUC)

Period: 02-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.8	17.1	4.7	7.7	21.7	7.3	8.8	21.5	13.8	36.4	12.1	36.9	1.4	310	0.492

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.4	84.3	67.5	80.9	65.8	77.8	64.6	68.6	81.8	66.8	78.8	65.2	76.1	7.1	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
63.6	97.1	76.0	62.2	92.3	73.5	60.9	88.0	71.3	34.4	82.3	32.9	79.2	31.5	76.3	73.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.8	10.1	9.0	DB	10.3	90.4	4.5	4.1	7.1	93.4	4.4	95.7	1.9	98.0	-1.4	101.0	(4)
(5)			WB	9.4	71.8	4.5	1.3	6.2	72.7	3.6	73.4	1.1	74.1	-2.2	75.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	30.7	33.1	40.8	48.7	56.3	62.4	65.7	64.2	57.1	49.0	39.7	32.0		
	DBStd	13.94	7.02	7.31	7.65	6.60	6.06	6.48	5.72	5.88	5.44	6.90	6.23	5.94		
	HDD50	2459	598	475	292	100	13	2	0	0	8	105	309	557		
	HDD65	6266	1062	893	749	489	278	126	63	87	242	497	758	1022		
	CDD50	1871	0	1	8	61	206	374	487	440	220	73	1	0		
	CDD65	205	0	0	0	0	7	48	85	61	4	0	0	0		
	CDH74	1631	0	0	0	5	88	395	664	455	24	0	0	0		
	CDH80	371	0	0	0	0	11	81	153	126	0	0	0	0		
(14) Wind	WSAvg	2.9	2.5	3.5	3.6	3.5	3.6	3.3	3.2	2.9	2.6	2.0	2.1	2.2		
(15) Precipitation	PrecAvg	49.1	2.4	2.1	2.8	3.2	4.3	5.2	5.8	5.7	4.5	5.2	5.4	2.7		
	PrecMax	66.5	9.8	9.1	5.3	5.9	8.6	8.2	11.3	8.5	7.7	14.7	15.7	7.3		
	PrecMin	39.3	0.4	0.4	0.4	0.9	1.6	2.9	1.9	2.8	1.3	0.3	0.3	0.2		
	PrecStd	7.0	1.9	1.7	1.3	1.2	1.6	1.5	2.0	1.5	1.7	3.3	4.1	1.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.0	58.8	66.3	72.8	81.8	85.9	88.0	89.7	77.2	70.3	55.8	47.4		
		MCWB	42.7	47.5	51.2	57.2	64.2	66.8	69.7	69.5	64.1	59.8	48.5	38.8		
	2%	DB	46.7	51.6	61.1	68.2	76.6	82.4	84.4	83.3	73.3	65.1	52.3	44.1		
		MCWB	40.1	42.4	49.1	54.9	61.7	65.9	68.0	67.8	62.2	56.2	46.5	39.3		
	5%	DB	42.4	46.6	57.0	64.7	72.5	78.9	81.0	78.8	69.9	61.8	50.1	41.7		
		MCWB	36.5	38.5	46.7	52.5	59.5	64.8	65.9	65.9	59.9	54.3	46.5	37.3		
	10%	DB	39.2	42.8	53.0	61.1	68.6	74.8	77.6	75.1	66.9	58.8	48.1	39.4		
		MCWB	34.6	35.9	44.0	50.5	56.9	62.4	64.7	63.8	58.2	53.1	45.2	35.6		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	47.6	51.9	57.9	65.1	68.9	71.0	70.8	65.2	60.0	50.3	43.1		
		MCDB	50.4	58.3	64.6	71.3	79.1	82.2	85.5	86.7	75.4	68.9	52.7	44.9		
	2%	WB	40.5	42.5	49.5	55.4	62.5	67.0	68.8	68.5	62.9	57.5	48.7	40.3		
		MCDB	45.7	50.7	60.3	67.0	75.2	79.8	82.3	81.4	71.3	63.3	50.8	42.8		
	5%	WB	37.7	39.1	47.0	53.1	60.1	65.3	66.9	66.5	60.9	55.4	47.3	38.2		
		MCDB	41.1	45.5	56.2	63.2	71.0	77.0	79.0	77.6	67.9	60.2	49.3	40.8		
	10%	WB	35.4	36.7	44.6	51.0	57.8	63.3	65.2	64.5	59.3	53.6	45.7	36.2		
		MCDB	38.2	41.7	51.7	59.9	67.3	73.4	76.0	73.7	65.5	58.3	47.6	38.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	12.3	16.7	17.2	18.6	18.3	19.4	17.6	16.3	14.0	10.1	9.0		
		MCDBR	14.1	18.4	23.7	23.8	24.8	25.3	24.5	24.0	20.4	18.3	13.3	11.8		
	5% WB	MCWBR	9.4	12.1	14.7	12.9	13.2	12.1	11.6	11.6	11.1	10.8	9.0	8.3		
		MCWBR	12.2	17.5	22.6	22.6	23.4	23.6	22.6	22.3	17.9	16.3	10.3	10.0		
(40) Clear-Sky Solar Irradiance	taub	taub	0.268	0.295	0.342	0.387	0.402	0.411	0.409	0.402	0.367	0.341	0.306	0.272		
		taud	2.373	2.300	2.260	2.238	2.265	2.283	2.293	2.330	2.402	2.449	2.478	2.449		
	Ebn at Noon	Ebn at Noon	264	277	278	275	274	271	270	267	267	258	248	250		
		Edh at Noon	22	30	36	41	41	41	40	37	31	26	21	19		
(44) All-Sky Solar Radiation	RadAvg	RadAvg	445	726	1063	1326	1536	1715	1749	1540	1144	776	446	372		
	RadStd	RadStd	51	95	129	173	187	154	146	165	141	89	80	54		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (85 neighbors)		+0.79	N/A	N/A	+1.21	+0.76	+0.59	N/A	-212	+137	+77

Nomenclature: See separate page